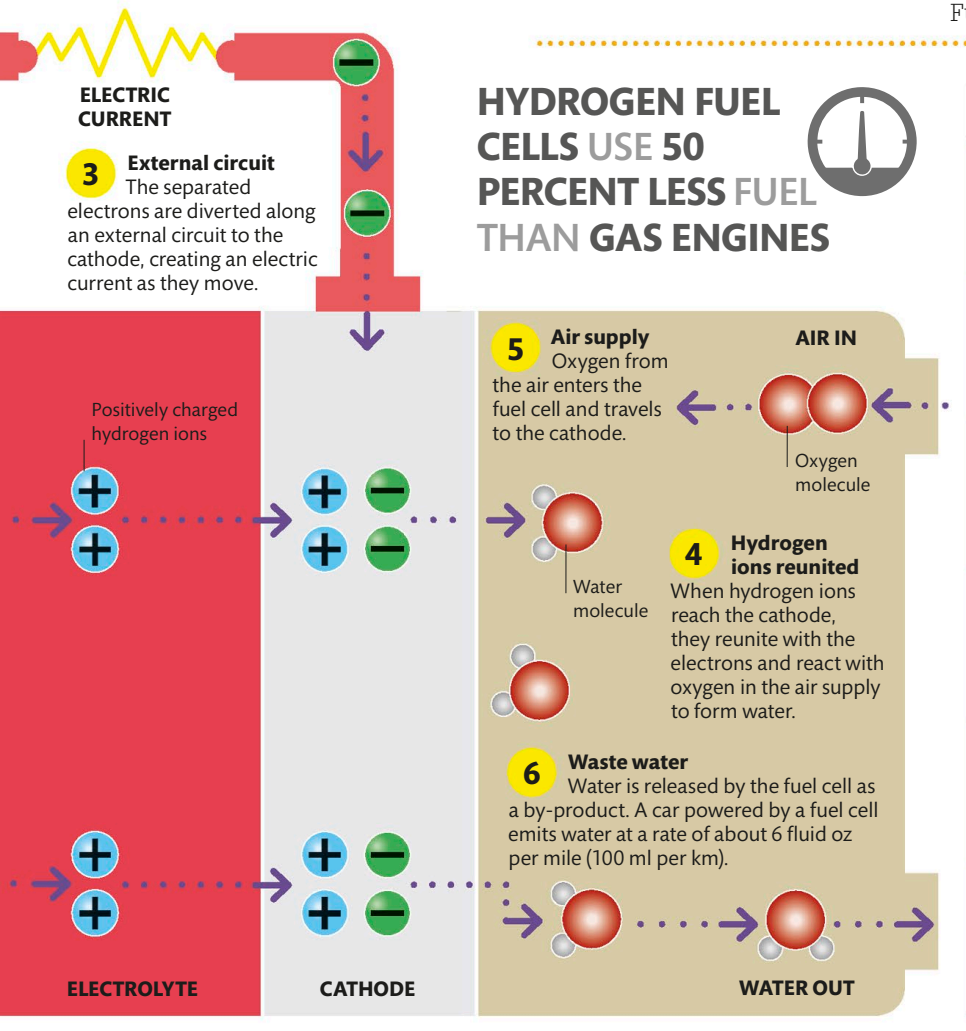
HYDROGEN FUEL CELLS USE 50 PERCENT LESS FUEL THAN GAS ENGINES



USES OF FUEL CELLS

Fuel cells remain an emerging technology but one with a vast range of potential applications that make use of its compact, convenient, and exhaust-free source of electricity.

Vehicles
Increasing numbers of fork-lift trucks, zero-emission buses, city trams, and some cars are powered by fuel cells.

Military
Small cells can power soldiers' electronic devices, while larger cells can keep drones in the air for long periods.

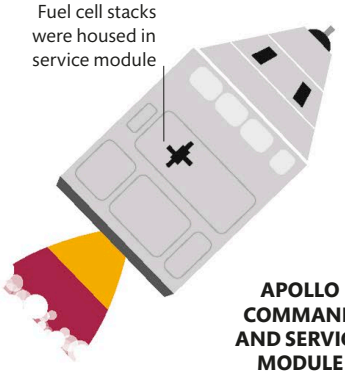
Portable electronics
Micro fuel cells are being developed to recharge smartphones, tablets, and other mobile devices.

Space
Fuel cells are a common source of power in spacecraft. Manned craft also use the fresh water they produce.

Aircraft
Experimental fuel-cell aircraft exist, but airliners are more likely to use them as back-up power supplies.

FUEL CELLS IN SPACE

Fuel cells first traveled into space on NASA's Gemini missions in 1965–1966. A stack of hydrogen cells housed in the service module also provided electrical power for the Apollo manned missions to the Moon (1969–1972). Each of the fuel cells contained 31 separate cells connected in series. The fuel cells used by Apollo proved highly successful, producing up to 2,300 watts of power while being less bulky than batteries and more efficient than solar panels.



ARE HYDROGEN FUEL CELLS SAFE?

Fears persist since hydrogen is extremely flammable, but fuel cells are manufactured with rigorous safeguards, and hydrogen tanks in vehicles are very tough and crushproof.